

GML_04_01-buns.R

frank

2024-05-06

```
# Variablen löschen und Pakete Laden-----
rm(list=ls(all=TRUE))

library(corrplot)

## Warning: Paket 'corrplot' wurde unter R Version 4.3.3 erstellt
## corrplot 0.92 loaded

library(lmtest)

## Warning: Paket 'lmtest' wurde unter R Version 4.3.3 erstellt
## Lade nötiges Paket: zoo
##
## Attache Paket: 'zoo'
## Die folgenden Objekte sind maskiert von 'package:base':
##
##      as.Date, as.Date.numeric

library(sandwich)

## Warning: Paket 'sandwich' wurde unter R Version 4.3.3 erstellt

library(car)

## Warning: Paket 'car' wurde unter R Version 4.3.3 erstellt
## Lade nötiges Paket: carData

library(tseries)

## Registered S3 method overwritten by 'quantmod':
##   method      from
## as.zoo.data.frame zoo

library(forecast)
library(stats)
library(fGarch)

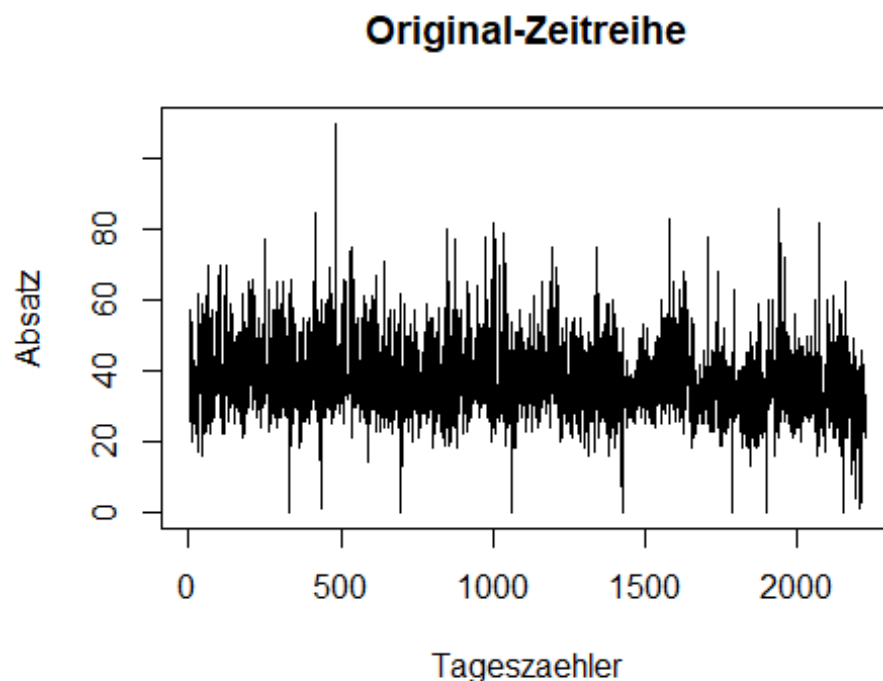
## NOTE: Packages 'fBasics', 'timeDate', and 'timeSeries' are no longer
## attached to the search() path when 'fGarch' is attached.
##
```

```
## If needed attach them yourself in your R script by e.g.,
##       require("timeSeries")

#Library(rugarch)
#Daten laden-----
daten = read.csv("GML_04_01 buns.csv")
weekday = daten[,5]

#Wochentage hinzuf?gen
mon = daten$mon <- ifelse(weekday == 0, 1, 0)
tue = daten$tue <- ifelse(weekday == 1, 1, 0)
wed = daten$wed <- ifelse(weekday == 2, 1, 0)
thu = daten$thu <- ifelse(weekday == 3, 1, 0)
fri = daten$fri <- ifelse(weekday == 4, 1, 0)
sat = daten$sat <- ifelse(weekday == 5, 1, 0)
sun = daten$sun <- ifelse(weekday == 6, 1, 0)

x11()
plot(daten$AMOUNT, type = "l", lty = 1, ylab = "Absatz", xlab =
"Tageszaehler", main = "Original-Zeitreihe")
```



```
#check pot time trend and corona impact, first Lockdown 22 Macrh 2020

daten$date[1879]
## [1] "3/22/2020"

tail(daten$date) #data ends in corona time
```

```
## [1] "2/27/2021" "2/28/2021" "3/1/2021" "3/2/2021" "3/3/2021" "3/4/2021"

daycounter = c(1:length(daten$AMOUNT))
covid = rep(0,length(daten$AMOUNT))
covid[1879:length(daten$AMOUNT)] = rep(1, (length(daten$AMOUNT)-1878))
mean(covid)#percentage of covid days

## [1] 0.1563342

summary(lm(daten$AMOUNT~daycounter+covid))

##
## Call:
## lm(formula = daten$AMOUNT ~ daycounter + covid)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37.492  -6.486  -2.282   3.982  72.804
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  38.1298196  0.4931076   77.326 < 2e-16 ***
## daycounter   -0.0019329  0.0004542   -4.255 2.18e-05 ***
## covid        -1.6309690  0.8037310   -2.029  0.0426 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.71 on 2223 degrees of freedom
## Multiple R-squared:  0.02404,    Adjusted R-squared:  0.02316
## F-statistic: 27.38 on 2 and 2223 DF,  p-value: 1.798e-12

#Mulitple Linear Regression-----

#Lin Reg. final
reg_final = lm(AMOUNT~ daycounter + mon + tue + wed + thu + fri + sat + sun +
               BOCHUM_TOTAL + KEMNADE_IN_FLAMMER + holiday + covid + -1 ,
data = daten)
summary(reg_final, digits = 4)

##
## Call:
## lm(formula = AMOUNT ~ daycounter + mon + tue + wed + thu + fri +
##      sat + sun + BOCHUM_TOTAL + KEMNADE_IN_FLAMMER + holiday +
##      covid + -1, data = daten)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -71.320  -4.318  -0.254   4.067  55.569
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## daycounter   -0.0018968  0.0003447   -5.503 4.17e-08 ***
```

```
## mon          35.3335914  0.5657842  62.451 < 2e-16 ***
## tue          33.5728103  0.5642086  59.504 < 2e-16 ***
## wed          34.3420027  0.5644066  60.846 < 2e-16 ***
## thu          35.1827653  0.5675384  61.992 < 2e-16 ***
## fri          34.1390956  0.5669721  60.213 < 2e-16 ***
## sat          37.4061440  0.5660995  66.077 < 2e-16 ***
## sun          52.4756284  0.5658694  92.735 < 2e-16 ***
## BOCHUM_TOTAL  3.4435418  1.8326925   1.879 0.06038 .
## KEMNADE_IN_FLAMMER 3.9289039  1.8494494   2.124 0.03375 *
## holiday      20.1647763  1.0848763  18.587 < 2e-16 ***
## covid        -1.6576039  0.6101629  -2.717 0.00665 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.123 on 2214 degrees of freedom
## Multiple R-squared:  0.9529, Adjusted R-squared:  0.9527
## F-statistic: 3733 on 12 and 2214 DF, p-value: < 2.2e-16
```

#Diagnostik zum Modell-----

#[A0]: Der Zusammenhang zwischen der abh?ngigen und den unabh?ngigen Variablen muss linear sein.

#RESET-Test

`resettest`(reg_final) *#H0: Kein Spezifikationserror.*

##

RESET test

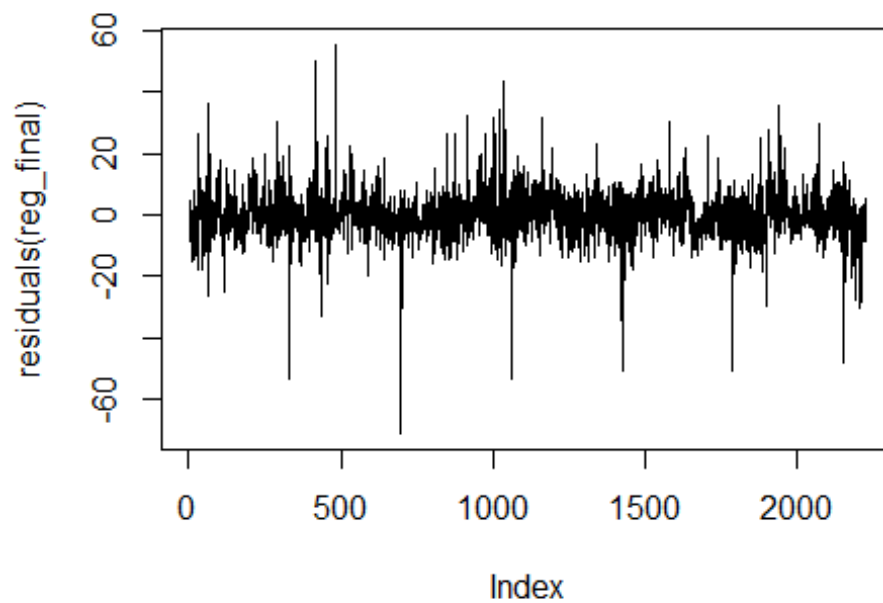
##

data: reg_final

RESET = 41.415, df1 = 2, df2 = 2212, p-value < 2.2e-16

#[A3]: Die Varianz der Residuen ist konstant und endlich (Homoskedastizit?t bzw.keine Heteroskedastizit?t)

`plot`(residuals(reg_final), type='l')

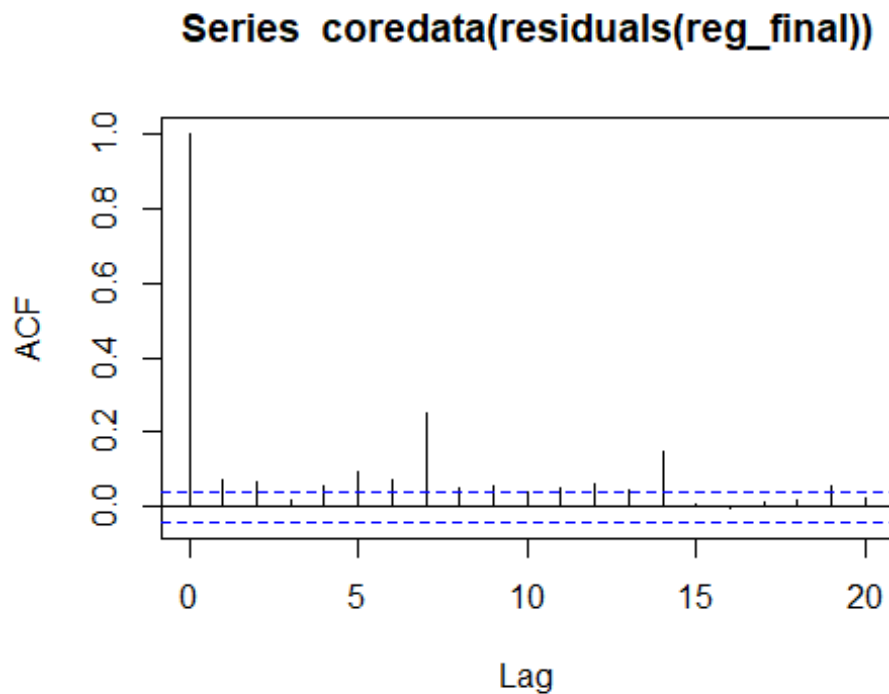


```
#Breusch-Pagan-Test
bptest(reg_final) #H0: Homoskedastizität.

##
## studentized Breusch-Pagan test
##
## data: reg_final
## BP = 381.75, df = 11, p-value < 2.2e-16

#Ergebnis: Heteroskedastizität liegt vor.

#[A4]: Die Residuen dürfen nicht untereinander korrelieren (keine
Autokorrelation).
#Korrelogramm
acf(coredata(residuals(reg_final)), ci = 0.95, lag.max = 20)
```



#Durbin-Watson-Test

`dwtest(reg_final)` *#H0: Keine Autokorrelation.*

##

Durbin-Watson test

##

data: reg_final

DW = 1.8607, p-value = 0.0004232

alternative hypothesis: true autocorrelation is greater than 0

#Ergebnis: Autokorrelation liegt vor.

#Da Heteroskedastizitaet und Autokorrelation vorliegen, muss mit HAC SE gearbeitet werden:

`coeftest(reg_final)#compare`

##

t test of coefficients:

##

	Estimate	Std. Error	t value	Pr(> t)	
## daycounter	-0.00189684	0.00034471	-5.5028	4.17e-08	***
## mon	35.33359141	0.56578417	62.4507	< 2.2e-16	***
## tue	33.57281027	0.56420862	59.5042	< 2.2e-16	***
## wed	34.34200271	0.56440657	60.8462	< 2.2e-16	***
## thu	35.18276529	0.56753844	61.9919	< 2.2e-16	***
## fri	34.13909557	0.56697212	60.2130	< 2.2e-16	***
## sat	37.40614402	0.56609955	66.0770	< 2.2e-16	***
## sun	52.47562837	0.56586943	92.7345	< 2.2e-16	***

```
## BOCHUM_TOTAL      3.44354181  1.83269248  1.8790  0.060382 .
## KEMNADE_IN_FLAMMER 3.92890389  1.84944940  2.1244  0.033750 *
## holiday           20.16477630  1.08487625 18.5872 < 2.2e-16 ***
## covid             -1.65760385  0.61016294 -2.7167  0.006646 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(reg_final,vcov=NeweyWest(reg_final))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## daycounter      -0.00189684  0.00049729 -3.8143 0.0001403 ***
## mon              35.33359141  0.73765012 47.9002 < 2.2e-16 ***
## tue              33.57281027  0.68648967 48.9050 < 2.2e-16 ***
## wed              34.34200271  0.64063042 53.6066 < 2.2e-16 ***
## thu              35.18276529  0.68619331 51.2724 < 2.2e-16 ***
## fri              34.13909557  0.69168672 49.3563 < 2.2e-16 ***
## sat              37.40614402  0.68797429 54.3714 < 2.2e-16 ***
## sun              52.47562837  0.78063815 67.2215 < 2.2e-16 ***
## BOCHUM_TOTAL      3.44354181  2.05394542  1.6765 0.0937717 .
## KEMNADE_IN_FLAMMER 3.92890389  2.36746915  1.6595 0.0971492 .
## holiday           20.16477630  3.44124370  5.8597 5.329e-09 ***
## covid             -1.65760385  0.92666501 -1.7888 0.0737862 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#[A5]: Es darf kein sehr starker linearer Zusammenhang zwischen den einzelnen erkl?renden Variablen bestehen

#[keine bzw. h?chstens geringe Multikollinearit?t).

#Variance Inflation Factor (VIF)

```
vif(reg_final)
```

```
## Warning in vif.default(reg_final): No intercept: vifs may not be sensible.
```

```
##          daycounter          mon          tue
wed
##          6.626112          1.542926          1.534345
1.535422
##          thu          fri          sat
sun
##          1.552509          1.549412          1.544647
1.543391
##          BOCHUM_TOTAL KEMNADE_IN_FLAMMER          holiday
covid
##          1.018182          1.036887          1.052516
1.963755
```

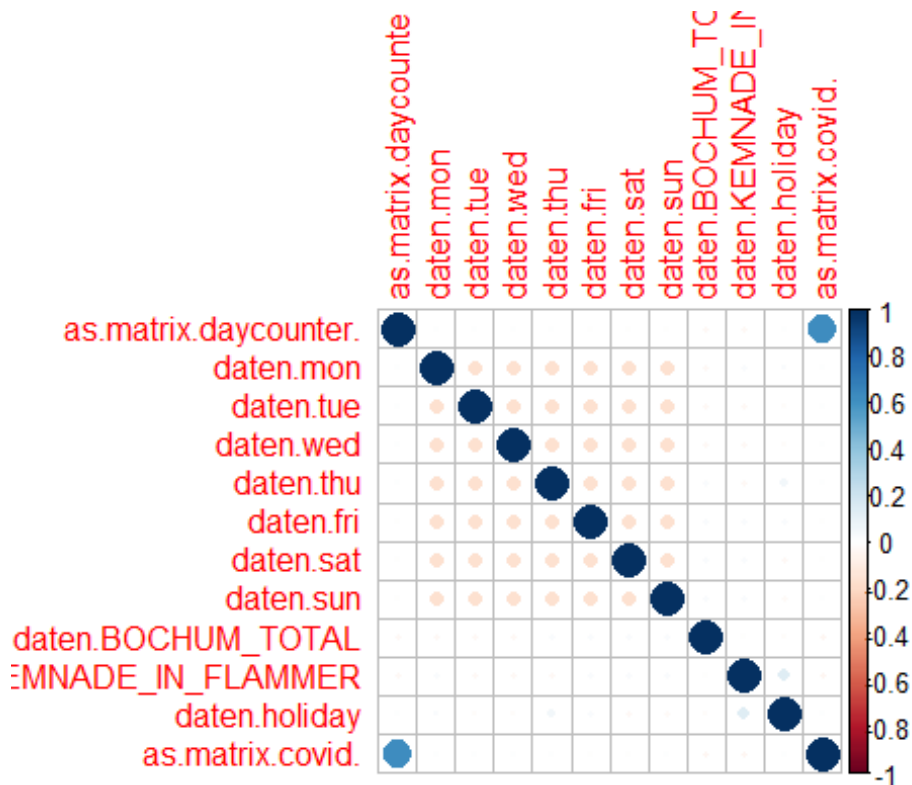
#Korrelationsmatrix

```
df <- data.frame(as.matrix(daycounter), daten$mon, daten$tue, daten$wed,
daten$thu, daten$fri, daten$sat, daten$sun,
```

```

        daten$BOCHUM_TOTAL, daten$KEMNADE_IN_FLAMMER, daten$holiday,
as.matrix(covid))
corrplot(cor(df))#does not help

```



```

#Zeitreihenanalyse Residuen-----
#acf shows structure

```

```

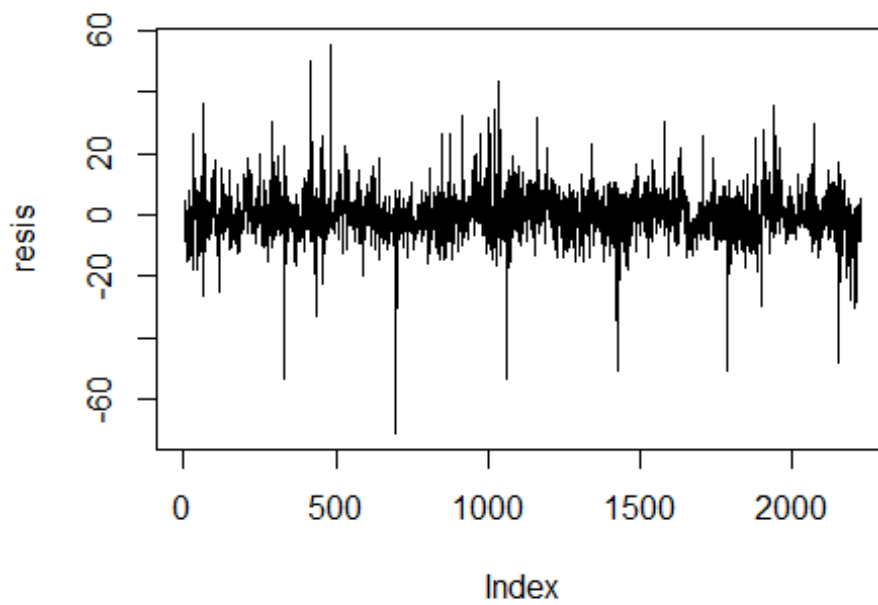
resis = residuals(reg_final)

```

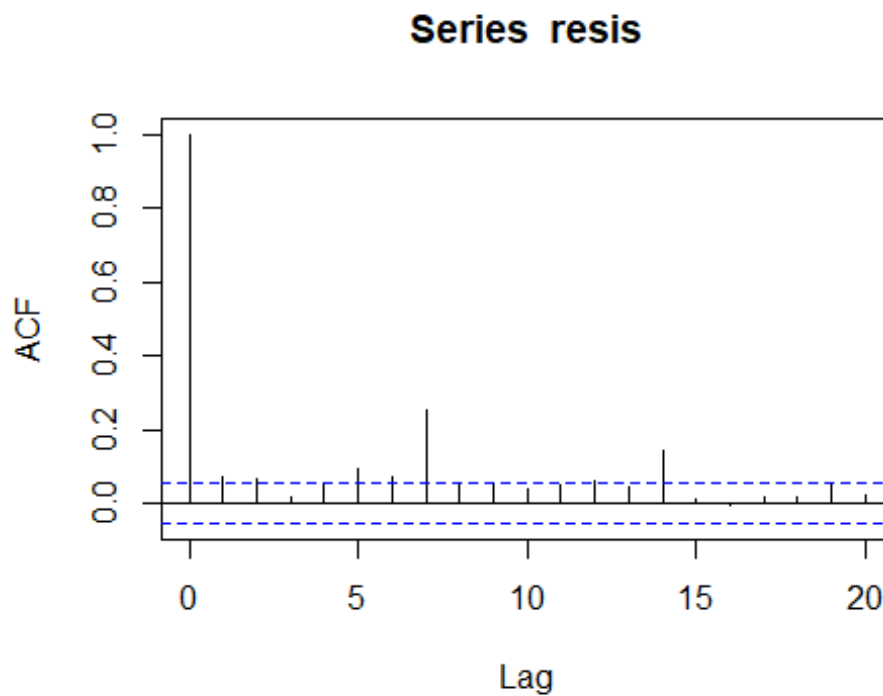
```

x11()
plot(resis, type='l')

```

```
x11()  
#Korrelogramm, für alpha = 1% diesmal weil viele Daten  
acf(resis, ci = 0.99, lag.max = 20)
```



```

#this will get clear ...
adf.test(resis)

## Warning in adf.test(resis): p-value smaller than printed p-value

##
## Augmented Dickey-Fuller Test
##
## data: resis
## Dickey-Fuller = -8.3202, Lag order = 13, p-value = 0.01
## alternative hypothesis: stationary

# Phillips-Perron (PP) Test
pp.test(resis)

## Warning in pp.test(resis): p-value smaller than printed p-value

##
## Phillips-Perron Unit Root Test
##
## data: resis
## Dickey-Fuller Z(alpha) = -2486.4, Truncation lag parameter = 8, p-value
## = 0.01
## alternative hypothesis: stationary

# Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test
kpss.test(resis)

## Warning in kpss.test(resis): p-value greater than printed p-value

##
## KPSS Test for Level Stationarity
##
## data: resis
## KPSS Level = 0.23956, Truncation lag parameter = 8, p-value = 0.1

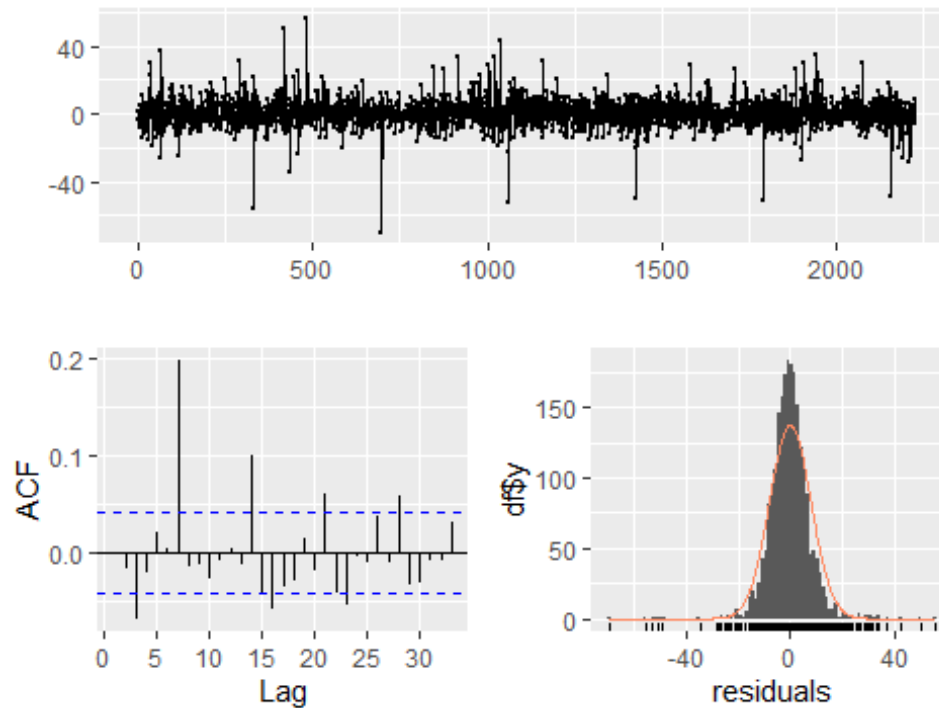
#see https://otexts.com/fpp2/arima-r.html
#variation of the Hyndman-Khandakar algorithm
#which combines unit root tests, minimisation of the AICc to obtain an ARIMA
model
autofitarima <- auto.arima(resis, max.p = 14, max.q = 14, stationary=TRUE)
autofitarima

## Series: resis
## ARIMA(2,0,1) with zero mean
##
## Coefficients:
##          ar1      ar2      ma1
##      0.9540  0.0142 -0.9186
## s.e.  0.0283  0.0227  0.0189
##
## sigma^2 = 63.34: log likelihood = -7774.47
## AIC=15556.94 AICc=15556.95 BIC=15579.77

```

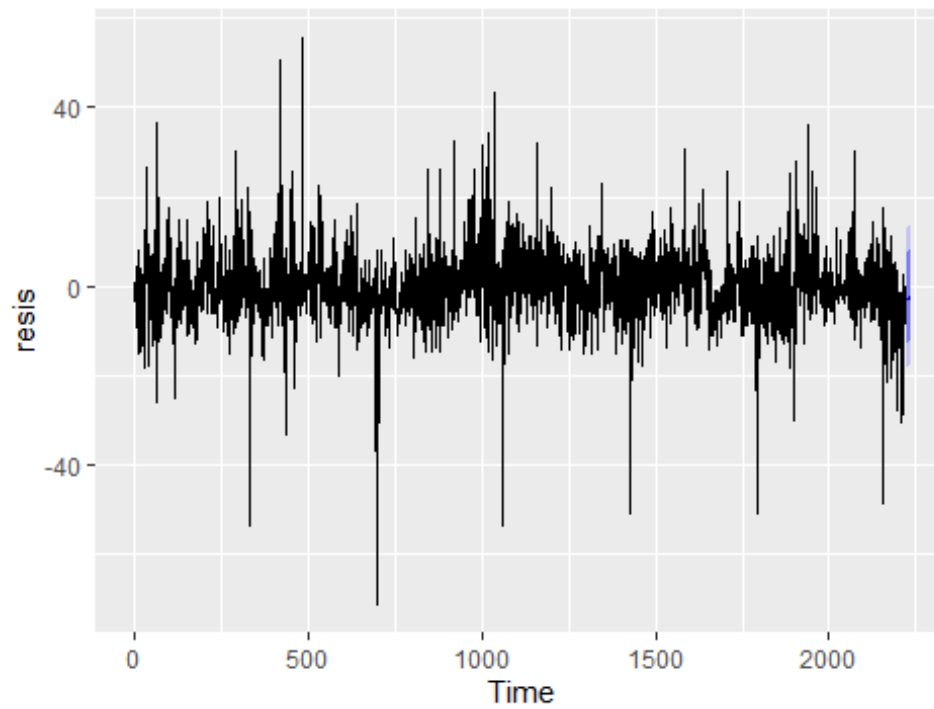
```
x11()
checkresiduals(autofitarima)#evident that GARCH needed
```

Residuals from ARIMA(2,0,1) with zero mean



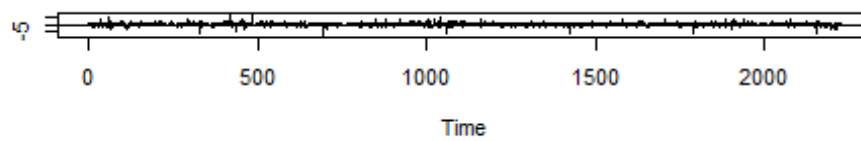
```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(2,0,1) with zero mean
## Q* = 103.43, df = 7, p-value < 2.2e-16
##
## Model df: 3.    Total lags used: 10
x11()
autoplot(forecast(autofitarima))
```

Forecasts from ARIMA(2,0,1) with zero mean

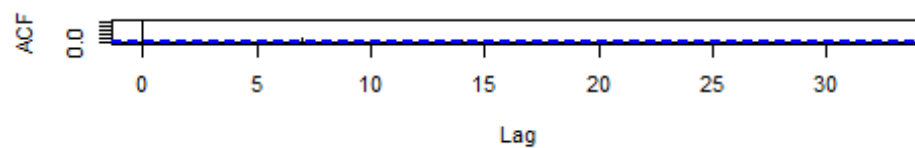


```
x11()  
tsdiag(autofitarima)
```

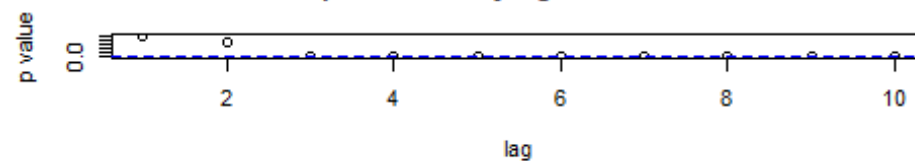
Standardized Residuals



ACF of Residuals



p values for Ljung-Box statistic



#deep dive

```
fitarma21 = arma(resis, order = c(2, 1), include.intercept = F)
```

```
summary(fitarma21)#note significant paras!
```

```
##
```

```
## Call:
```

```
## arma(x = resis, order = c(2, 1), include.intercept = F)
```

```
##
```

```
## Model:
```

```
## ARMA(2,1)
```

```
##
```

```
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -69.867  -4.110  -0.176   4.020  55.707
```

```
##
```

```
## Coefficient(s):
```

```
##      Estimate Std. Error t value Pr(>|t|)
## ar1    0.95435    0.02793   34.17  <2e-16 ***
## ar2    0.01382    0.02265    0.61    0.542
## ma1   -0.91937    0.01833  -50.15  <2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Fit:
```

```
## sigma^2 estimated as 63.28, Conditional Sum-of-Squares = 140682.1, AIC = 15555.77
```

```
fitarma11 = arma(resis, order = c(1, 1), include.intercept = F)
```

```
summary(fitarma11)#small change in para values, sign same, enjoy inside unit circle
```

```
##
```

```
## Call:
```

```
## arma(x = resis, order = c(1, 1), include.intercept = F)
```

```
##
```

```
## Model:
```

```
## ARMA(1,1)
```

```
##
```

```
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -69.8426  -4.1481  -0.1672   3.9916  55.6820
```

```
##
```

```
## Coefficient(s):
```

```
##      Estimate Std. Error t value Pr(>|t|)
## ar1    0.96903    0.01138   85.16  <2e-16 ***
## ma1   -0.92079    0.01742  -52.87  <2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Fit:
```

```
## sigma^2 estimated as 63.36, Conditional Sum-of-Squares = 140912.7, AIC = 15556.4
```

```

#cross chk forecast. of ARMA11
fitarma11$fitted.values[100]

##      100
## 1.104654

resis[100]

##      100
## 9.78354

#hence error100

error100 = resis[100]-fitarma11$fitted.values[100]
error100

##      100
## 8.678886

forecast101 = 0.96903 * resis[100] -0.92079 * error100
forecast101

##      100
## 1.489112

fitarma11$fitted.values[101]

##      101
## 1.489113

#end of chking, go for garch

# GARCH(1,1) - Normal

# Modell 1 - Normal GARCH
NormalGgf<-garchFit(~garch(1,1),data=resis,include.mean = F, trace=FALSE)
coef(NormalGgf)

##      omega      alpha1      beta1
## 15.31631609  0.04834394  0.72096099

summary(NormalGgf)

##
## Title:
##  GARCH Modelling
##
## Call:
##  garchFit(formula = ~garch(1, 1), data = resis, include.mean = F,
##    trace = FALSE)
##
## Mean and Variance Equation:
##  data ~ garch(1, 1)
## <environment: 0x0000020742274828>

```

```

## [data = resis]
##
## Conditional Distribution:
## norm
##
## Coefficient(s):
##      omega      alpha1      beta1
## 15.316316    0.048344    0.720961
##
## Std. Errors:
## based on Hessian
##
## Error Analysis:
##      Estimate Std. Error t value Pr(>|t|)
## omega    15.31632     4.43185   3.456 0.000548 ***
## alpha1    0.04834     0.01721   2.809 0.004974 **
## beta1     0.72096     0.07613   9.471 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log Likelihood:
## -7805.754      normalized: -3.506628
##
## Description:
## Mon May  6 09:37:10 2024 by user: frank
##
##
## Standardised Residuals Tests:
##
##      Jarque-Bera Test  R      Chi^2 8698.199 0
##      Shapiro-Wilk Test  R      W    0.9143244 0
##      Ljung-Box Test    R      Q(10) 213.3141 0
##      Ljung-Box Test    R      Q(15) 281.5604 0
##      Ljung-Box Test    R      Q(20) 291.2525 0
##      Ljung-Box Test    R^2    Q(10) 15.30894 0.121198
##      Ljung-Box Test    R^2    Q(15) 18.41426 0.2415134
##      Ljung-Box Test    R^2    Q(20) 21.57838 0.3638198
##      LM Arch Test      R      TR^2 15.84307 0.19853
##
## Information Criterion Statistics:
##      AIC      BIC      SIC      HQIC
## 7.015952 7.023644 7.015948 7.018761

#JB Tests (#H0 normal)of normalized resis point to misspecified distribtuion
#Ljung checks independence of resis and sqrd resis,
#looks good, can keep H0: Indep

# GARCH(1,1) - skew & more fat

# hints at potential developmts
SkewedStdGgf<-garchFit(~garch(1,1),data=resis,include.mean =

```

```

FALSE,cond.dist="sstd", trace=FALSE)
SkewedStdGgf

##
## Title:
## GARCH Modelling
##
## Call:
## garchFit(formula = ~garch(1, 1), data = resis, cond.dist = "sstd",
## include.mean = FALSE, trace = FALSE)
##
## Mean and Variance Equation:
## data ~ garch(1, 1)
## <environment: 0x000002073ed7b448>
## [data = resis]
##
## Conditional Distribution:
## sstd
##
## Coefficient(s):
## omega alpha1 beta1 skew shape
## 6.50046 0.01897 0.87820 1.08128 3.84740
##
## Std. Errors:
## based on Hessian
##
## Error Analysis:
## Estimate Std. Error t value Pr(>|t|)
## omega 6.50046 5.03151 1.292 0.196
## alpha1 0.01897 0.01265 1.500 0.134
## beta1 0.87820 0.08796 9.984 <2e-16 ***
## skew 1.08128 0.02769 39.056 <2e-16 ***
## shape 3.84740 0.31129 12.360 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log Likelihood:
## -7558.425 normalized: -3.395519
##
## Description:
## Mon May 6 09:37:11 2024 by user: frank

#include lagged Y
head(cbind(daten$AMOUNT[-1],daten$AMOUNT[-length(daten$AMOUNT)]))

## [,1] [,2]
## [1,] 34 35
## [2,] 57 34
## [3,] 32 57
## [4,] 27 32
## [5,] 26 27
## [6,] 30 26

```



```

#Lin Reg. w Lagged Y
reg_final_extended = lm(AMOUNT[-1] ~ AMOUNT[-length(AMOUNT)] + daycounter[-1]
+ mon[-1] + tue[-1] + wed[-1] + thu[-1] + fri[-1] + sat[-1]
+ sun[-1] +
BOCHUM_TOTAL[-1] + KEMNADE_IN_FLAMMER[-1] + holiday[-1] +
covid[-1] + -1, data = daten)
summary(reg_final, digits = 4)

##
## Call:
## lm(formula = AMOUNT ~ daycounter + mon + tue + wed + thu + fri +
##     sat + sun + BOCHUM_TOTAL + KEMNADE_IN_FLAMMER + holiday +
##     covid + -1, data = daten)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -71.320  -4.318  -0.254   4.067  55.569
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## daycounter      -0.0018968   0.0003447  -5.503 4.17e-08 ***
## mon              35.3335914   0.5657842  62.451 < 2e-16 ***
## tue              33.5728103   0.5642086  59.504 < 2e-16 ***
## wed              34.3420027   0.5644066  60.846 < 2e-16 ***
## thu              35.1827653   0.5675384  61.992 < 2e-16 ***
## fri              34.1390956   0.5669721  60.213 < 2e-16 ***
## sat              37.4061440   0.5660995  66.077 < 2e-16 ***
## sun              52.4756284   0.5658694  92.735 < 2e-16 ***
## BOCHUM_TOTAL      3.4435418   1.8326925   1.879  0.06038 .
## KEMNADE_IN_FLAMMER 3.9289039   1.8494494   2.124  0.03375 *
## holiday           20.1647763   1.0848763  18.587 < 2e-16 ***
## covid            -1.6576039   0.6101629  -2.717  0.00665 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.123 on 2214 degrees of freedom
## Multiple R-squared:  0.9529, Adjusted R-squared:  0.9527
## F-statistic: 3733 on 12 and 2214 DF, p-value: < 2.2e-16

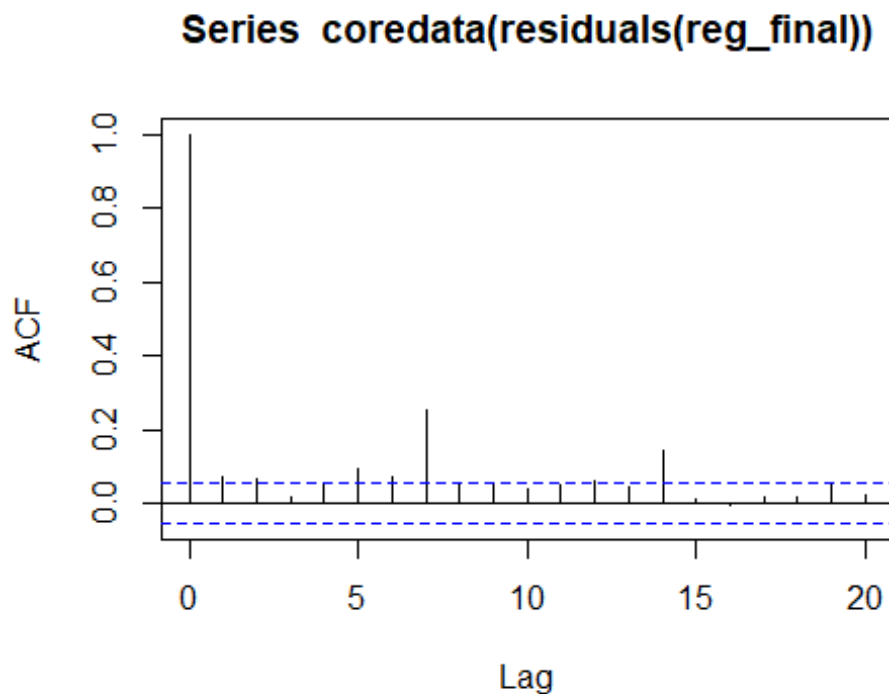
summary(reg_final_extended, digits = 4)

##
## Call:
## lm(formula = AMOUNT[-1] ~ AMOUNT[-length(AMOUNT)] + daycounter[-1] +
##     mon[-1] + tue[-1] + wed[-1] + thu[-1] + fri[-1] + sat[-1] +
##     sun[-1] + BOCHUM_TOTAL[-1] + KEMNADE_IN_FLAMMER[-1] + holiday[-1] +
##     covid[-1] + -1, data = daten)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -71.488  -4.213  -0.316   4.040  55.063
##

```

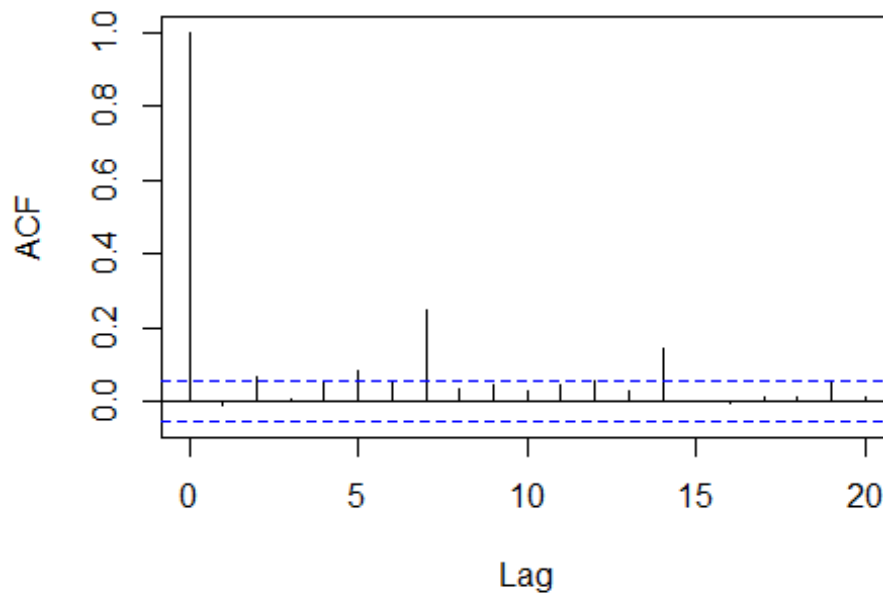
```
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## AMOUNT[-length(AMOUNT)]  0.075745  0.019732   3.839 0.000127 ***
## daycounter[-1]          -0.001752  0.000346  -5.064 4.44e-07 ***
## mon[-1]                 31.310928  1.189897  26.314 < 2e-16 ***
## tue[-1]                 30.829721  0.909292  33.905 < 2e-16 ***
## wed[-1]                 31.760977  0.876646  36.230 < 2e-16 ***
## thu[-1]                 32.531673  0.892690  36.442 < 2e-16 ***
## fri[-1]                 31.377099  0.915053  34.290 < 2e-16 ***
## sat[-1]                 34.748895  0.893023  38.912 < 2e-16 ***
## sun[-1]                 49.607578  0.936075  52.995 < 2e-16 ***
## BOCHUM_TOTAL[-1]        3.409286  1.827480   1.866 0.062234 .
## KEMNADE_IN_FLAMMER[-1]  3.866158  1.844238   2.096 0.036165 *
## holiday[-1]             20.600866  1.087712  18.940 < 2e-16 ***
## covid[-1]               -1.532874  0.609395  -2.515 0.011960 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.099 on 2212 degrees of freedom
## Multiple R-squared:  0.9532, Adjusted R-squared:  0.9529
## F-statistic: 3466 on 13 and 2212 DF, p-value: < 2.2e-16

#[A4]: Die Residuen dürfen nicht untereinander korrelieren (keine Autokorrelation).
acf(coredata(residuals(reg_final)), ci = 0.99, lag.max = 20)
```



```
acf(coredata(residuals(reg_final_extended)), ci = 0.99, lag.max = 20)
```

Series coredata(residuals(reg_final_extended))



```
#Durbin-Watson-Test
```

```
dwtest(reg_final_extended) #H0: Keine Autokorrelation.
```

```
##
```

```
## Durbin-Watson test
```

```
##
```

```
## data: reg_final_extended
```

```
## DW = 2.0188, p-value = 0.6505
```

```
## alternative hypothesis: true autocorrelation is greater than 0
```

```
dwtest(reg_final)
```

```
##
```

```
## Durbin-Watson test
```

```
##
```

```
## data: reg_final
```

```
## DW = 1.8607, p-value = 0.0004232
```

```
## alternative hypothesis: true autocorrelation is greater than 0
```

```
#Ergebnis: Autokorrelation ist weg.
```

```
#Breusch-Pagan-Test
```

```
bptest(reg_final_extended) #H0: Homoskedastizität.
```

```
##
```

```
## studentized Breusch-Pagan test
```

```
##
```

```
## data: reg_final_extended
```

```
## BP = 379.02, df = 12, p-value < 2.2e-16
```

#Ergebnis: Heterosekdastizitaet liegt vor.

#deshalb:

#manchmal muss man ausprobieren mit den Ans?tzen bis es l?uft numerisch (-> R Hilfe!), oder Wechsel auf

`coeftest`(reg_final_extended)

```
##
## t test of coefficients:
##
##               Estimate Std. Error t value Pr(>|t|)
## AMOUNT[-length(AMOUNT)]  0.07574490  0.01973236  3.8386 0.0001272 ***
## daycounter[-1]          -0.00175227  0.00034601 -5.0643 4.438e-07 ***
## mon[-1]                 31.31092828  1.18989743 26.3140 < 2.2e-16 ***
## tue[-1]                 30.82972087  0.90929183 33.9052 < 2.2e-16 ***
## wed[-1]                 31.76097715  0.87664587 36.2301 < 2.2e-16 ***
## thu[-1]                 32.53167289  0.89268998 36.4423 < 2.2e-16 ***
## fri[-1]                 31.37709856  0.91505329 34.2899 < 2.2e-16 ***
## sat[-1]                 34.74889535  0.89302352 38.9115 < 2.2e-16 ***
## sun[-1]                 49.60757761  0.93607478 52.9953 < 2.2e-16 ***
## BOCHUM_TOTAL[-1]         3.40928576  1.82748024  1.8656 0.0622343 .
## KEMNADE_IN_FLAMMER[-1]   3.86615827  1.84423783  2.0963 0.0361650 *
## holiday[-1]              20.60086591  1.08771244 18.9396 < 2.2e-16 ***
## covid[-1]                -1.53287446  0.60939466 -2.5154 0.0119599 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

`coeftest`(reg_final_extended,vcov = `vcovHC`(reg_final_extended, type = "HC0"))*#White estimator*

```
##
## t test of coefficients:
##
##               Estimate Std. Error t value Pr(>|t|)
## AMOUNT[-length(AMOUNT)]  0.07574490  0.02027224  3.7364 0.0001914 ***
## daycounter[-1]          -0.00175227  0.00034025 -5.1500 2.836e-07 ***
## mon[-1]                 31.31092828  1.32663338 23.6018 < 2.2e-16 ***
## tue[-1]                 30.82972087  0.96519626 31.9414 < 2.2e-16 ***
## wed[-1]                 31.76097715  0.81417643 39.0099 < 2.2e-16 ***
## thu[-1]                 32.53167289  0.89661794 36.2826 < 2.2e-16 ***
## fri[-1]                 31.37709856  0.89482424 35.0651 < 2.2e-16 ***
## sat[-1]                 34.74889535  0.87619083 39.6590 < 2.2e-16 ***
## sun[-1]                 49.60757761  1.00588341 49.3174 < 2.2e-16 ***
## BOCHUM_TOTAL[-1]         3.40928576  1.84807900  1.8448 0.0652042 .
## KEMNADE_IN_FLAMMER[-1]   3.86615827  2.14673533  1.8009 0.0718473 .
## holiday[-1]              20.60086591  3.28954279  6.2625 4.537e-10 ***
## covid[-1]                -1.53287446  0.59529023 -2.5750 0.0100883 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Ende